

BED

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Backend Development (BED)

Diploma in IT
Year 2 (2026/26), Semester 1

Module Synopsis

- **This module provides students with the fundamental knowledge and skills required in back-end development. It will also cover the integration of back-end system with databases, possibly external systems as well as consuming data from web application protocol interfaces (APIs).**

What is a Backend Development?

- Working on the **Server Side Software**
- Focuses on everything you can't see on a website – **databases, back-end logic, application programming interface (APIs), architecture and server**

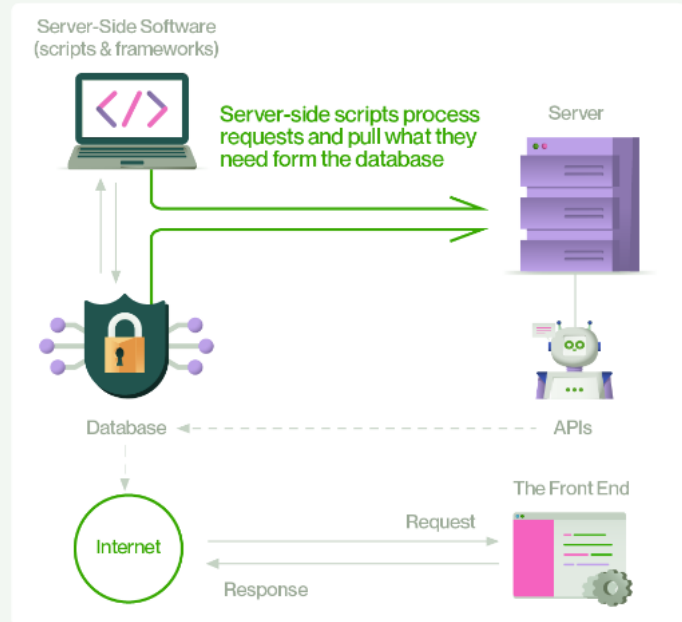
Back-End Development & Frameworks in Server-Side Software

upwork

Frameworks are libraries of server-side programming languages that construct the back-end structure of a site

The “**stack**” comprises the database, server-side framework, server, and operating system (OS).

APIs structure how data is exchanged between a database and any software accessing it.



(Courtesy of Upwork Global Inc., extracted from <https://www.upwork.com/resources/beginners-guide-back-end-development>)

Module Objectives

At the end of this semester, you should be able to:

- **Implement Backend API using Node.js**
- **Integrate Backend API with a Frontend application and a database to perform CRUD (create, read, update & delete) operations**
- **Evaluate advanced features like security, testing and integrating third-party libraries.**

Core Programming Modules

Diploma in Information Technology

Year 1

1.1 Programming I

1.2 Programming II

1.2 Databases

1.2 Front-End Development

Year 2

2.1 Back-End Development

2.1 Software Project Management

2.2 Full Stack Development Project

Assessments



- **Continual Assessment (CA) – 15%**
 - **Weekly Practical – 10%**
 - **Weekly Quiz (via OAL content) – 5%**
- **Common Test (Wk 8) – 30%**
 - **Topics from Wk 1 – 7**
- **Assignment 1 (Wk 11 – 17) – 40%**
 - **Work in teams of 4 to 5**
- **Summative Quiz (Wk 15) – 15%**
 - **Topics from Wk 1 – 13**

Lesson Plan – Wk 1 to 8

Wk	Begin Date	Topics
1	20 Apr	Introduction to Backend Development - Explore common backend technologies - Front-end technologies vs Back-end technologies - Introduction to Node.js and Express.js - Install Node.js and npm - Setup a basic Express.js application
2	27 Apr *	Building API using Node.js and Express - Introduction to API - Retrieve data using GET Request - Create data using POST Request - Update data using PUT Request - Delete data using DELETE Request - Accessing API using Postman
3	04 May	Building API with Database Integration - Connecting to a relational database - Creating records in database - Retrieving records from database - Updating records in database - Deleting records from database
4	11 May	Building Maintainable & Secure APIs - Model-View-Controller (MVC) architecture - Server-side input data validation using middleware - Error handling using Try-Catch-Finally construct - Parameterized queries to prevent SQL Injections
5	18 May	Whitespace Week - Integrating back-end APIs to front-back application through JavaScript API calls.
6	25 May *	Managing Entities and Relationships in APIs - Construct complex SQLs with JOINS to retrieve JASON ouput data compiled from multiple tables
7	01 Jun *	Common Test Revision
8	08 Jun	Common Test

Lesson Plan – Wk 11 to 17

11	29 Jun	User Authentication and Authorization - Introduction of JSON Web Token (JWT) - JWT generation during User Authentication - API Authorization with JWT
12	06 Jul	User Authentication and Authorization - API Authorization with JWT Token
13	13 Jul	Unit Testing & API Documentation - Implement unit tests with Jest framework for API controllers and models, including mocking dependencies - Generate API documentation with Swagger to understand input data requirement for testing of API end-points from web browser.
14	20 Jul	Whitespace Week Assignment Consultation
15	27 Jul	Summative Quiz and Assignment Consultation
16	03 Aug	Assignment Demo
17	10 Aug *	Assignment Demo
18	17 Aug	
19	24 Aug	

*** Holiday:**

01 May - Labour Day

27 May - Hari Raya Haji

01 Jun - Vesak Day

10 Aug - National Day

Learning Journey



- **Practical teaching approach**
 - You will learn Backend Development concepts through hands-on exercises. (Note: 10% marks penalty for late submission of 1 week or more.)
- **Flip-Classroom approach**
 - Come to class prepared—complete the OAL content and lecture materials beforehand to make the most of in-class activities.
- **Self-Directed learning**
 - Take control of your own learning journey. Tutors will help with the facilitation.

Software Version Control



- Use **Github** (<https://github.com>) for version control and collaboration during software development.
 - Refer to the [Linkedin Learning](#) course : “[Learning GitHub by Emanuele Bartolesi](#)”
- Create Github repository for Practical and named the repository as: **BED2026Apr_Practical_P<your_Class>_<your_Student ID>**
 - e.g.: **BED2026Apr_Practical_P01_s12345678**
 - Add your **tutor's e-mail address** to the repository with “**Collaborator**” access rights (Settings > Collaborators). (tutor's email address can be found in POLITEMall > Our Tutors)
- Push your Practical work to the Git Repository regularly and on time along with any Practical submission in BrightSpace (if applicable).

Tools

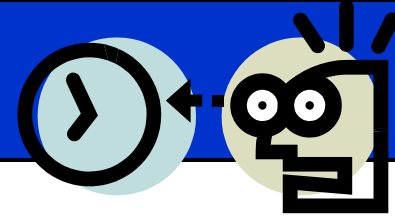
- **Node.js & Express**
- **Git & Github**
- **Microsoft Visual Studio Code**
- **Postman Native Desktop App**
- **Microsoft SQL Server Express**
- **Microsoft SQL Server Management Studio**

Class Attendance



- ATTENDANCE is **COMPULSORY** for ALL CLASSES.
- Medical Certificate (MC) must be submitted online in NPal2 Student within **48 hours**, from time of absence for classes/assessment. **Late** submission will not be accepted.
- If the absence is for Class Test, Common Test, Exams and other assessment, the original MC/supporting document must be submitted to ICT Admin Office within 2 working days.

Class Attendance



- Be **PUNCTUAL** for class.
 - You will be marked as **ABSENT** if you arrive **LATE** for lesson without valid reason.
- Your grade may be **capped at 'D'** if you fails to register a minimum of 70% attendance for a module.

Classroom Rules

**Let us build a
Conducive Learning Environment
together!**



- Handphone set to SILENT mode in class.
 - You may ask for tutor's permission to answer VERY URGENT CALL outside the classroom.
- No drinking & eating in classroom.
- No playing of computer games during class.
- No vulgar language.



End of Briefing

**Remember to
have fun!**

